Paper Solutions - index

Doc	Paper	Style	Status in your exam
P1	2024 final (Sales, SQLite, PyTorch)	pandas + charts + DB + NN, 40 marks	**MAIN TEMPLATE for the mock**
(T3)	Practice Exam Task 3 (Height logger)	CSV append, numpy, histogram	likely PyTorch replacement; solved in `Topics/T3_Files_CSV_Exceptions` and `Solutions_py/practice_task3_height_logger.py`
P2	2021 Registration	DB + validation	GUI removed; logic reusable (console version provided)
P3	2022 Bus fare	DB + fare rule	GUI removed; logic reusable
P4	2023 Library + "ChatGPT"	DB + validation + text parsing	GUI removed; logic reusable

All runnable solutions live in `Practice/Solutions_py/`:

File	What
`2024_sales_sqlite_pytorch_solution.py`	the full 2024 paper (tested; re-runnable)
`2024_twist_drills.py`	11 "what if the question changes" variations
`practice_task3_height_logger.py`	Practice Task 3
`2021_registration_users_solution.py` + `_console.py`	2021 (window / terminal)
`2022_bus_fare_travelprice_solution.py` + `_console.py`	2022
`2023_library_librarian_solution.py` + `_console.py`	2023

How to study a paper solution (do not just read it): 1. Read the paper's wording and cover the solution. Write your own plan as comments. 2. Write the code from your plan. Run it. 3. Compare with the solution; for each difference ask "is mine wrong, or just different?" 4. Read the line-by-line explanation for every line you could not write alone. 5. Retype the solution the next day from the explanations only.

Findings that go beyond the official solution (all tested; explained in P1): - the official 2024 solution under-reports 2023 sales by £50 (blank Price filled with 0), plots all years instead of 2023, and does not prevent duplicates (re-running doubles the DB); - `sales_data.csv` contains 6 exact duplicate rows.